GAMEHUB

FINAL RELEASE NOTES

Project	GameHub	**Release**	Final Version
Platform	Android	**Repository**	https://github.com/ST10345327/Loot-Gamehub

Project Team

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1. Release Overview

GameHub has grown from a prototype with basic game search and details into a game discovery and management application. This release adds authentication, personal game collections, game comparison, random discovery, customisable settings, notifications, and limited offline access.

2. Features

2.1 Authentication and Account Management

- User registration and login with email and password.
- Input validation and error handling.
- Password hashing on the backend.
- Session management to keep users signed in.
- Google Sign-In using Firebase Authentication and Android Credential Manager.
- Backend verification of Firebase ID tokens.
- Google account identification using unique Google UIDs.
- Protection against automatically linking Google sign-in to an existing password-protected account.
- Google Sign-In tested successfully.

2.2 Game Search and Discovery

- IGDB API integration for game information.
- Game search and discovery.
- Game details including titles, descriptions, release dates, genres, platforms, ratings, and artwork where available.

2.3 Personal Game Management

- Personal game library, favourites, and wishlist.
- Ability to move games from the wishlist into the library.

2.4 Game Comparison and Random Discovery

- Game comparison for selected titles.
- Surprise Me feature for random game discovery.

2.5 Settings and Personalisation

- Light, Dark, and System theme options.
- English, Setswana, and isiZulu language preferences.
- Preferences are saved and retained between app sessions.

2.6 Notifications

- Wishlist-related and game-release notifications.
- Backend notification-generation functionality.
- Protected backend endpoints for scheduled notification generation.
- Notification delivery tested successfully.

2.7 Offline Functionality

- The Settings section can be accessed and used offline.
- Offline access is limited to Settings. Other features, including authentication, game discovery, library and wishlist data, and notifications, require an internet connection.

2.8 Backend and Security

- Node.js and Express backend with MySQL database integration and migration support.
- Authentication and protected API endpoints.
- Firebase Admin SDK verification for Google Sign-In.
- Backend validation and error handling.
- Automated tests for authentication and other backend functionality.

3. Testing and Quality Assurance

The final development report recorded the following results:

Test / Verification	Recorded result
Backend tests	38 tests passed
Android unit tests	86 unit tests passed
Android debug build	Completed successfully
Physical-device testing	Installed and launched on a physical Android phone
Google Sign-In	Tested successfully
Notifications	Tested successfully
Offline Settings access	Tested and confirmed to work offline

Note: The test counts and verification results above are based on the final development report provided by the project team.

4. Final Summary

The final version of GameHub expands the original game-search prototype with email and Google authentication, game discovery, a personal library, favourites, a wishlist, game comparison, random discovery, customisable settings, notifications, and limited offline access.

Offline functionality is specifically limited to Settings; the rest of the app depends on online services. The Android app and backend work together to provide authentication, game data, user collections, and notifications.